End User, enterprise and cloud computing

End user computing

- End-user computing (EUC) is a combination of technologies, policies, and processes that gives your workforce secure, remote access to applications, desktops, and data they need to get their work done.
- Modern enterprises use EUC so that their employees can work from wherever they are, across multiple devices, in a safe and scalable way.
- EUC program gives users immediate access to the digital technologies they need for productivity, both on-premises and remotely in the cloud.

Benefits of End user computing

- **Business continuity**
  1. Hundreds of millions of people now work remotely, with no or limited access to company hardware.
  2. EUC is critical to business continuity as employees access the business applications and data they need on their personal devices.

Benefits of End user computing contd/

- **Security**
  1. EUC services and applications are centrally managed. They store data and desktops centrally in the cloud and stream only pixels out to endpoint devices, avoiding sensitive data downloads to user devices.
  2. End users can use multi-factor authentication to access EUC systems, even further decreasing the risk of unauthorised data access if an employee's device is lost or stolen.

Benefits of End user computing contd/

- **Agility**
  1. Cloud-based EUC solutions greatly improve agility by providing instant scalability, making it more straightforward for organisations to deploy and manage secure virtual desktop infrastructure.
  2. You can release new software and new tools or turn existing Windows- or Linux-based applications into Software as a Service (SaaS) applications faster than you could on premises.
  3. EUC also makes it more efficient to oversee migrations and deployments.

Benefits of End user computing contd/

- **Cost savings**
  1. It is more cost effective for enterprises to manage centralised desktops and EUC solutions in the cloud.
  2. You can manage and install applications in a single location, reducing management complexity and cost.
  3. You can also add and remove users on demand and pay only for what you use, when you use it.
  4. You no longer need to overbuy hardware and licenses for peak user capacity.

Benefits of End user computing contd/

- **Testing**
  1. EUC is an optimum environment for rapid prototyping and testing. Organisations can offer new tools to their workers to test how effective and user friendly they are on a large scale.
  2. EUC allows you to change design and structure to rapidly carry out testing campaigns.

Benefits of End user computing contd/

- **Collaboration**
  1. Teams using EUC services can securely collaborate on projects, with their content stored on the cloud.
  2. You can also share project information with external users for cross-organisational collaboration with changes made in real time.

What are EUC use cases?

- **Remote workforce**
  1. Is used to empower people to work from wherever they are based and from multiple devices.
  2. You can use EUC services to onboard new workers remotely, and grant access to the databases and applications they need to do their jobs effectively, all while keeping organisational data secure and compliant.

What are EUC use cases? contd/

- **Contact centers**
  1. Call center staff no longer have to be in physical call centers to do their jobs.
  2. Virtual contact centers run in the cloud allow agents to access virtual desktops and applications.
  3. EUC ensures that sensitive corporate information remains on the cloud rather than devices used by contact agents.

What are EUC use cases? contd/

- **Trials, demos, and training**
  1. With EUC services, you can provide software trials on demand with no local installs or special hardware required.
  2. Similarly, sales teams can deliver your company's software technologies for demos on any device. Training and certification workshops for your software can occur online, with no need for a physical lab or classroom.

What are the technologies used in EUC? Contd/

- **Virtual Desktop Infrastructure (VDI)**
  1. Virtual desktop infrastructure (VDI) is a software solution that allows you to provision and manage individual desktop computing environments for end users from a centralised location.
  2. Virtual desktops are software applications that mimic real desktop computers.
  3. VDI technology creates a virtual desktop on a central server, and remote users can access this desktop from any physical machine over the internet.

What are the technologies used in EUC? Contd/

- **DaaS**
  1. Desktop as a Service (DaaS) is a cloud computing service where providers deliver virtual desktops to end users online.
  2. It is set up with a per-user subscription license.
  3. Service providers oversee the management of maintenance, backup, data storage, and updates.
  4. DaaS offers advantages over VDI because it doesn't require infrastructure setup and management costs and avoids overprovisioning.

What are the technologies used in EUC? Contd/

- **Application as a service**
  1. Application as a service technology lets you make your Windows- and Linux-based applications available to users without rewriting the code.
  2. You can quickly and easily install, test, and update your applications and set scaling parameters to meet the needs of your end users.

Enterprise Computing

- Enterprise computing refers to the information technology infrastructure, systems, and applications that businesses and large organisations use to manage and process their data on a large scale.
- It encompasses a wide range of technologies and practices that support the core operations of an enterprise.

Key Components of Enterprise

- **Computing**: Hardware: Servers, storage devices, networking equipment, and other physical components.
- **Software**: Operating systems, database management systems, application software, and middleware.
- **Data**: Information assets that are crucial to the organisation's operations.
- **Network**: The infrastructure that connects devices and systems within the enterprise.
- **Security**: Measures to protect data and systems from unauthorized access and threats.

Common Enterprise Computing Systems

- Enterprise Resource Planning (ERP): Integrates various business processes, including finance, human resources, and supply chain management.
- Customer Relationship Management (CRM): Manages interactions with customers and prospects.
- Supply Chain Management (SCM): Optimizes the flow of goods and services from suppliers to customers.

Common Enterprise Computing Systems

- Business Intelligence (BI): Provides insights into data to support decision-making.
- Cloud Computing: Delivers computing services over the internet, such as infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS).

Benefits of Enterprise Computing

- Improved efficiency and productivity: Streamlines processes and reduces manual tasks.
- Enhanced decision-making: Provides data-driven insights to support strategic choices.
- Increased scalability: Enables organisations to grow and adapt to changing business needs.
- Improved customer satisfaction: Offers better products and services through efficient operations.
- Enhanced security: Protects sensitive data and systems from threats.

Cloud Computing

- Cloud computing is a general term for the delivery of hosted computing services and IT resources over the internet with pay-as-you-go pricing.
- Users can obtain technology services such as processing power, storage and databases from a cloud provider, eliminating the need for purchasing, operating and maintaining on-premises physical data centers and servers.

Cloud Computing

- A cloud can be private, public or a hybrid.
- A public cloud sells services to anyone on the internet.
- A private cloud is a proprietary network or a data center that supplies hosted services to a limited number of people, with certain access and permissions settings.
- A hybrid cloud offers a mixed computing environment where data and resources can be shared between both public and private clouds.

Cloud Computing

- The goal of cloud computing is to provide easy, scalable access to computing resources and IT services.
- Cloud infrastructure involves the hardware and software components required for the proper deployment of a cloud computing model.
- Cloud computing can also be thought of as utility computing or on-demand computing.

How does cloud computing work

- Cloud computing lets client devices access rented computing resources, such as data, analytics and cloud applications over the internet.
- Relies on a network of remote data centers, servers and storage systems that are owned and operated by cloud service providers.
- Providers are responsible for ensuring the storage capacity, security and computing power needed to maintain the data users send to the cloud.

How does cloud computing work

- The following steps are involved in cloud computing:
  1. An internet network connection links the front end -- the accessing client device, browser, network and cloud software applications -- with the backend, which consists of databases, servers, operating systems and computers.
  2. The backend functions as a repository, storing data accessed by the frontend.
  3. A central server manages communications between the front and back ends. It relies on protocols to facilitate the exchange of data. The central server uses both software and middleware to manage connectivity between different client devices and cloud servers.

How does cloud computing work

4. Typically, there's a dedicated server for each application or workload.
5. Cloud computing relies heavily on virtualization and automation technologies. Virtualization lets IT organisations create virtual instances of servers, storage and other resources that let multiple VMs or cloud environments run on a single physical server using software known as a hypervisor.

Types of cloud computing services

- Cloud services can be classified into three general service delivery categories and these are:

1. Infrastructure as a service (IaaS)
2. Platform as a service (PaaS
3. Software as a service (SaaS)

1. Infrastructure as service (IaaS)

- IaaS providers, such as Amazon Web Services (AWS), supply a virtual server instance and storage, as well as application programming interfaces (APIs) that let users migrate workloads to a virtual machine (VM).
- Users have an allocated storage capacity and can start, stop, access and configure the VM and storage as desired.
- IaaS providers offer small, medium, large, extra-large and memory- or compute-optimised instances, in addition to enabling customization of instances for various workload needs.

2. Platform as service (PaaS)

- In the PaaS model, cloud providers host development tools on their infrastructures.
- Users access these tools over the internet using APIs, web portals or gateway software.
- PaaS is used for general software development and many PaaS providers host the software after it is developed.
- Examples of PaaS products include Salesforce Lightning, AWS Elastic Beanstalk and Google App Engine.

3. Software as service (SaaS)

- SaaS is a distribution model that delivers software applications over the internet; these applications are often called web services.
- Users can access SaaS applications and services from any location using a computer or mobile device that has internet access.
- In the SaaS model, users gain access to application software and databases.
- An example of a SaaS application is Microsoft 365 for productivity and email services.